FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Main), 2015

HALF COURSE TEST – V

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 02 and 09 to 30)** contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (03 to 08) contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
(Single Correct Choice Type)

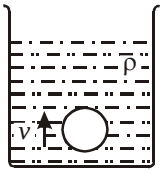
This section contains ***30 multiple choice questions***. Each question has four choice (A), (B), (C) and (D) out of which ***ONLY ONE*** is correct.

1. Kepler's third law of planetary motion when stated for circular orbits, has the form (v = orbital velocity of a planet and R = distance from the sun)

(A) $v \propto R$ (B) $v \propto 1/R$

(C) $v^2 \propto 1/R$ (D) none of the above

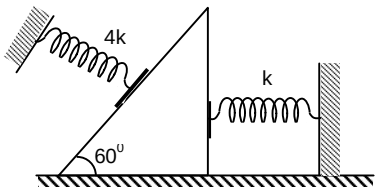
2. An air bubble of radius r rises steadily through a liquid of density ρ at the rate of *v*. Neglecting density of air, the coefficient of viscosity of liquid is



(A) $\frac{2}{9}\frac{r^2\rho g}{v}$ (B) $\frac{1}{3}\frac{r^2\rho g}{v}$

(C) $\frac{1}{9}\frac{r^2\rho g}{v}$ (D) None of these

3. A smooth wedge of mass m and angle of inclination 60^0 rests unattached between two springs of spring constant k and 4k, on a smooth horizontal plane, both springs in the unextended position. The time period of small oscillations of the wedge (assuming that the springs are constrained to get compressed along their length) equals



(A) $\pi\left(1+\frac{1}{2}\right)\sqrt{\frac{m}{k}}$ (B) $\pi\left(1+\frac{1}{\sqrt{3}}\right)\sqrt{\frac{m}{k}}$

(C) $2\pi\sqrt{\frac{m}{k}}$ (D) none of the above

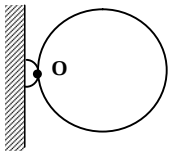
4. Two waveforms, $\psi_1(x, t) = A\sin(kx - \omega t)$ and $\psi_2(x, t) = A\sin(kx + \omega t + \pi/3)$, travelling along the x-axis, are superposed. The position of the nodes is given by ($k = \frac{2\pi}{\lambda}$, $\omega = 2\pi f$)

(A) $x_n = (n + 1/6)\frac{\lambda}{2}$ (B) $x_n = \left(n-\frac{1}{6}\right)\frac{\lambda}{2}$

(C) $x_n = n\frac{\lambda}{2}$ (D) none of these

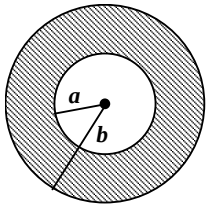
Space for rough work

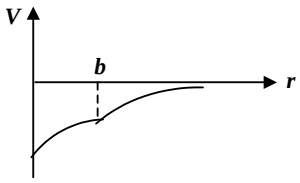
5. A uniform ring of mass m and radius r is hinged at point O so that it can rotate about horizontal axis passing through O in vertical plane. If the system is released from rest, then the initial acceleration of the ring is [Ring is in the vertical plane.]

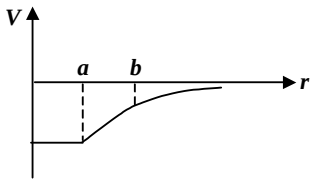


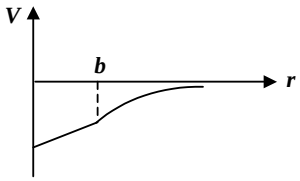
(A) g (B) g/2
(C) g/4 (D) g/8

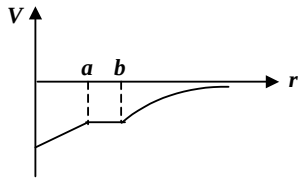
6. A sphere of mass *M* and radius *b* has a concentric cavity of radius *a* as shown in figure. The graph showing variation of gravitational potential *V* with distance *r* from the center of sphere is



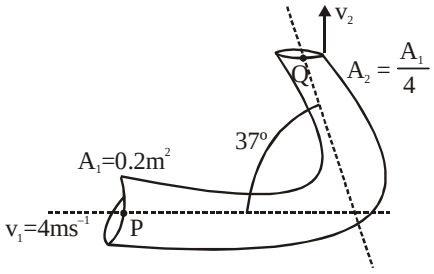
(A)


(B)


(C)


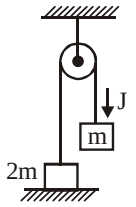
(D)


7. Oil enters the bend of a pipe in the horizontal plane with velocity 4 ms^{-1} and pressure 280 × 10^3 Nm^{-2} as shown in the figure. The pressure of oil at the point Q is 86λ. Find the value of λ. (KNm^{-2}).
(Take specific gravity of oil as 0.9 and sin 37º = 0.6)
(A) 1 (B) 2
(C) 3 (D) 4



Space for rough work

8. In the given figure an impulse J is given to the block of mass m in the downward direction. As a result of the impulse



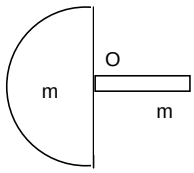
(A) both the blocks start moving with speed $\frac{J}{3m}$ in opposite directions

(B) both the blocks start moving with speed $\frac{J}{m}$ in opposite directions

(C) the centre of mass of the system is moving downwards

(D) the centre of mass of the system is not moving

9. A rod of mass m and length 2r is attached to the centre of a semi-circular lamina of equal mass m and radius r, such that it remains in the plane of the lamina. The moment of inertia of the system, about an axis passing through the centre O, perpendicular to the plane of the lamina equals



(A) $\frac{19}{6}mr^2$ (B) $\frac{5}{6}\ mr^2$

(C) $\frac{11}{6}\ mr^2$ (D) none of the above

10. A small ball of density ρ plunges into a liquid of density 2ρ, after falling freely from a height h above its surface. The motion of the particle will be (assuming no loss of energy and no splashing of liquid)

(A) simple harmonic with a period of $4\sqrt{\frac{2h}{g}}$ (B) simple harmonic with a period $2\sqrt{\frac{2h}{g}}$

(C) periodic with a time period $2\sqrt{\frac{2h}{g}}$ (D) periodic with a time period $4\sqrt{\frac{2h}{g}}$

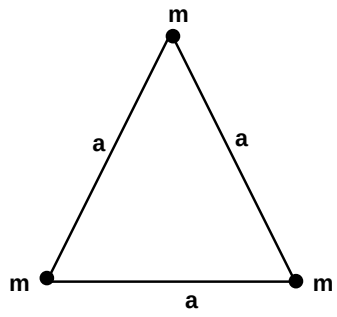
11. An ideal gas is taken through a process where dQ = dU - dW. The specific heat of the ideal gas in this process is (dQ = heat supplied to the gas, dU = change in internal energy, dW = work done by gas)

(A) C_v (B) $C_v + \frac{3}{2}R$

(C) $C_v - R$ (D) the process is impossible

Space for rough work

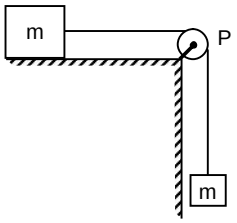
12. If the three point masses are released simultaneously to move under influence of mutual gravitational attraction, speed of each particle when each one of them has moved a distance $\left(\frac{a}{2\sqrt{3}}\right)$



(A) $\sqrt{\frac{6\sqrt{3}Gm}{\left(2-\sqrt{3}\right)a}}$ (B) $\sqrt{\frac{2Gm}{a}}$

(C) $\sqrt{\frac{Gm}{a}}$ (D) None of these

13. If the pulley 'P' as shown in the given arrangement is massless and there is no friction, force exerted by the pulley 'P' on the clamp would be



(A) $\sqrt{2}$ mg (B) $\frac{mg}{\sqrt{2}}$

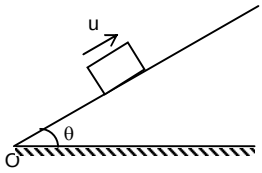
(C) 2mg (D) $\frac{mg}{2}$

14. A block of mass M is performing SHM with amplitude A on a smooth horizontal surface. At the extreme position a small block of mass m falls vertically and sticks to M. New amplitude of oscillation will be

(A) A (B) $A\sqrt{\frac{M}{M+m}}$

(C) $A\frac{M}{M+m}$ (D) $A\frac{M+m}{m}$

15. A block can slide down the incline plane with constant velocity. If it is projected with a velocity 'u' up the inclined plane at t = 0. The displacement of particle after a time t = $\frac{u}{g\sin\theta}$ from O will be

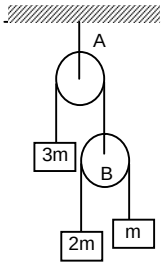


(A) $\frac{u^2}{4g\cos\theta}$ (B) $\frac{u^2}{2g\sin\theta}$

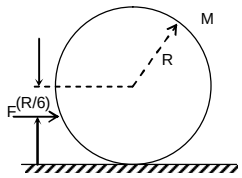
(C) zero (D) $\frac{u^2}{4g\sin\theta}$

Space for rough work

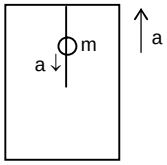
16. Consider the system shown when pulleys and strings are massless. If the system is released from rest, then centre of mass of the system (3m, 2m and m)
(A) is always at rest
(B) accelerates upwards
(C) accelerates downwards.
(D) has acceleration depending on the value of m.



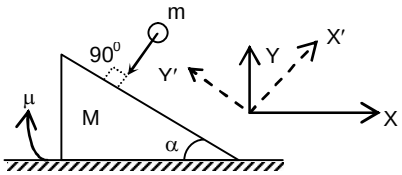
17. A solid sphere of mass M and radius R is placed vertically on a rough horizontal surface. If a horizontal force F is applied at a distance R/6 below the centre line parallel to the diameter of the sphere, the direction of frictional force will be
(A) forward (B) backward
(C) depends upon value of F (D) insufficient information.



18. A cabin is moving vertically upwards with an acceleration a. A massless string is attached to the ceiling of the cabin on which a bead of mass m is sliding downwards w.r.t. string with an acceleration a. The tension in the string is
(A) m(g + 2a) (B) m (g + a)
(C) mg (D) m (g – a)

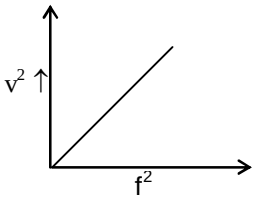


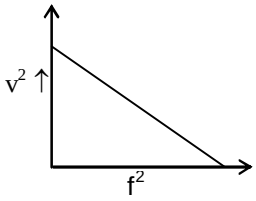
19. A ball of mass m strikes the inclined face of the wedge perpendicularly with speed v_0. The wedge is at rest on a rough horizontal surface before collision. The conservation of momentum is applicable for
(i) m as system, along y'
(ii) M as system, along y'
(iii) (M + m) as system, along x
(iv) (M + m) as system, along y
which of the following is correct ?
(A) (i) only (B) (i) + (ii) only
(C) (iii) only (D) (iii) + (iv) only.

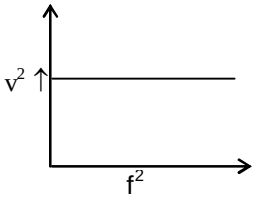


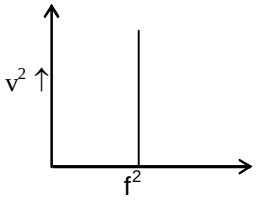
Space for rough work

20. A mass m is performing linear simple harmonic motion, then graph for square of acceleration f^2 and corresponding square of linear velocity v^2 is

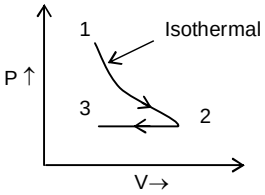
(A)


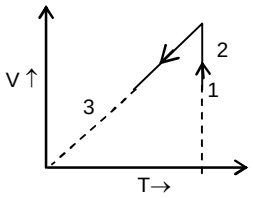
(B)


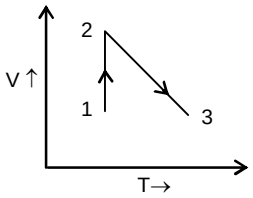
(C)


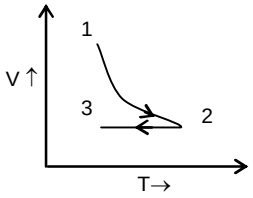
(D)


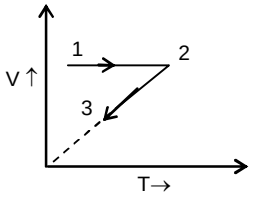
21. One mole of a monoatomic gas undergoes the processes 1 – 2 and 2 – 3 as shown. The corresponding graph for the above process for 2 moles of the gas is



(A)


(B)


(C)


(D)


Space for rough work

22. If ℓ_1 and ℓ_2 ($> \ell_1$) are the lengths of an air column for the first and second resonance when a tunning fork of frequency n is sounded on a resonance tube, then the minimum distance of the anti-node from the top end of the resonance tube is

(A) 2 (ℓ_2 - ℓ_1) (B) ½ (2ℓ_1 - ℓ_2)

(C) $\frac{\ell_2 - 3\ell_1}{2}$ (D) $\frac{\ell_2 + \ell_1}{2}$

23. Two different masses m and 2m of same ideal gas are heated separately in vessels of equal volume. The T-P curve for mass 2m makes angle α with T-axis and that for mass m makes angle β with T-axis then

(A) tan α = tan β (B) tan α = 2 tanβ

(C) tan β = 2 tanα (D) None of these

24. A viscous liquid flows through a uniform tube. As we move from the axis of the tube towards its wall, the speed of the liquid relative to the tube

(A) increases from zero to maximum value.

(B) decreases from maximum value to zero

(C) remains same throughout the cross section.

(D) increases from zero to a maximum value on one side and decreases from a maximum value to zero on the other side.

25. The displacement y of a particle executing a certain periodic motion is given by $y = 4\cos^2\left(\frac{1}{2}t\right)$ sin (1000t). This expression may be considered to be the superposition of n independent harmonic motions. Then, n is equal to

(A) 2 (B) 3

(C) 4 (D) 5

26. The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it from the surface. When the body reaches the centre of the planet, its speed will be

(A) v_e (B) $V_e\sqrt{2}$

(C) $\frac{v_e}{2}$ (D) zero

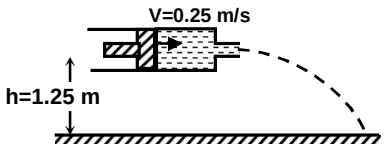
27. In a resonance tube, closed at one end by a smooth moving piston and the other end open, exhibits the first three resonance lengths, L_1, L_2 and L_3 for the same tuning fork. Then they are related by

(A) $L_3 = 2L_2 = 4\ L_1$ (B) $L_3 = \frac{5}{3}\ L_2 = 5\ L_1$

(C) $(L_3 - L_2) = (L_2 - L_1)$ (D) $(L_3 - L_2) = 2(L_2 - L_1)$

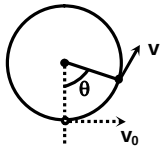
Space for rough work

28. A syringe of diameter D = 8 mm and having a nozzle of diameter d = 2 mm is placed horizontally at a height of 1.25 m as shown in the figure. An incompressible and non-viscous liquid is filled in syringe and the piston is moved at speed of 0.25 m/s. Find the range of liquid jet on the ground.



(A) 1 m (B) 4 m
(C) 3 m (D) 2 m

29. A pendulum of length ℓ = 1 m having a bob of mass m = 1 kg is hanging from a rigid support. If the bob is projected horizontally with a velocity $v_0 = \sqrt{35}$ m/s. The tension in the string is 6k Newton when angle made by string is 60° from vertical as shown. Find the value of k.



(A) 6 (B) 3
(C) 5 (D) 1

30. A particle of mass m moves towards a smooth vertical wall with a speed u (relative to the ground) and collides elastically with the wall; the wall moving towards the particle with a speed 2u (also relative to the ground). Assuming that the wall is extremely massive, the impulse delivered to the particle equals
(A) 5 mu (B) 6 mu
(C) 4 mu (D) none of these.

Space for rough work

Chemistry

PART – II

SECTION – A
(Single Correct Choice Type)

This section contains ***30 multiple choice questions***. Each question has four choice (A), (B), (C) and (D) out of which ***ONLY ONE*** is correct.

1. If $pK_a = 4$ and $K_a = Cx^2$ then van't Hoff factor for weak monobasic acid when C = 0.01 M is:
   (A) 1.01 (B) 1.02
   (C) 1.10 (D) 1.20

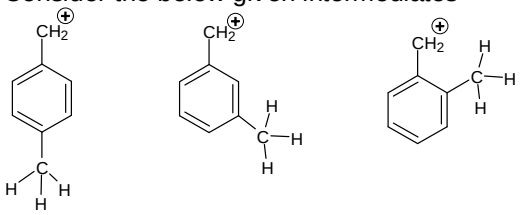
2. 10 g $CaCO_3$ is completely decomposed to X and CaO. X is passed into an aqueous solution containing one mole of sodium carbonate. What is the number of moles of sodium bicarbonate formed?
   (A) 0.2 (B) 0.1
   (C) 0.01 (D) 10

3. The average molar mass of air is 28.8. An open vessel at 27°C is heated upto t°C until 1/3rd of the air measured at final temperature escapes out. The rms velocity of air molecules at t°C is
   (A) 340 m sec^{-1} (B) 420 m sec^{-1}
   (C) 515 m sec^{-1} (D) 588 m sec^{-1}

4. Two solid compounds 'X' and 'Y' dissociates at certain temperature as follows :
   $X(s) \rightleftharpoons A(g) + 2B(g)$ $Kp_1 = 9 \times 10^{-3}$ atm^3
   $Y(s) \rightleftharpoons 2B(g) + C(g)$ $Kp_2 = 4.5 \times 10^{-3}$ atm^3.
   The total pressure(in atm) of gases over a mixture of 'X' and 'Y' is
   (A) 1.5 (B) 0.45
   (C) 0.6 (D) 0.9

Space for rough work

5. Consider the below given intermediates



(i) (ii) (iii)

The stability order of above intermediate are

(A) i > ii > ii (B) ii > i > iii
(C) iii > i > ii (D) iii = i > ii

6. At 550 K, the K_c for the following reaction is 10^4 mol^{-1} L.

$X(g) + Y(g) \rightleftharpoons Z(g)$

At equilibrium, it was observed that

$[X] = \frac{1}{2}[Y] = \frac{1}{2}[Z]$

What is the value of [Z] at equilibrium?

(A) 2×10^{-4} mol L^{-1}. (B) 10^{-4} mol L^{-1}.
(C) 2×10^{4} mol L^{-1}. (D) 10^{4} mol L^{-1}.

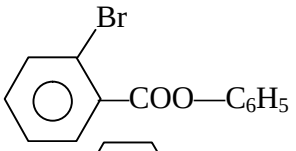
7. The predominant product formed when 3–methyl–2–pentene reacts with chlorine water is

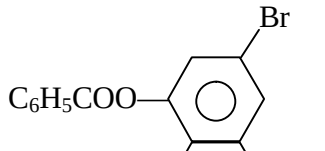
(A) $CH_3CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{Cl}{|}}{C}}-CH(OH)CH_3$ (B) $CH_3CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{Cl}{|}}{C}}-\underset{Cl}{\underset{|}{CH}}(OH)CH_3$

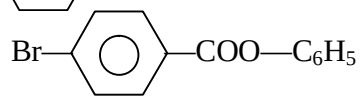
(C) $CH_3CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-\underset{Cl}{\underset{|}{CH}}-CH_3$ (D) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\underset{OH}{\underset{|}{CH}}-CH_3$

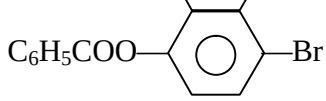
Space for rough work

8. The major product expected from the monobromination of phenyl benzoate is

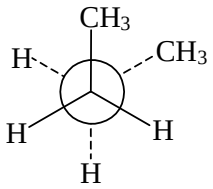
(A) 

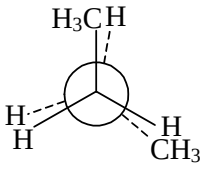
(B) 

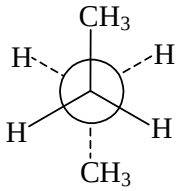
(C) 

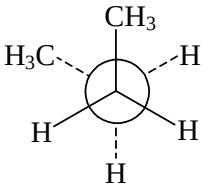
(D) 

9. Which of the following conformer of n–butane is achiral?

(A) 

(B) 

(C) 

(D) 

10. The correct order of basic strength the among the following is

(A) $Li_2O < Na_2O < K_2O < Rb_2O$ (B) $Rb_2O > Na_2O = K_2O > Li_2O$

(C) $Li_2O > Na_2O > K_2O > Rb_2O$ (D) $Na_2O > Li_2O > K_2O > Rb_2O$

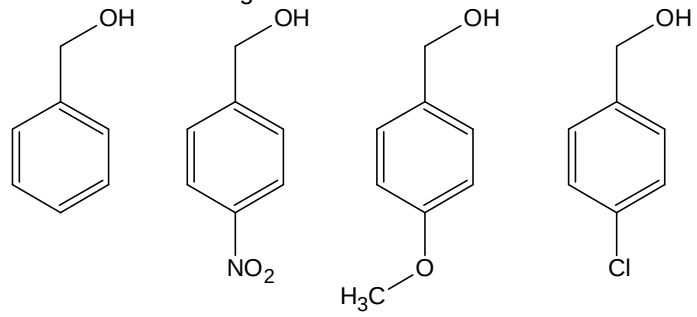
11. A hydrogen like ion in excited state can emit 3 spectral lines and the minimum energy emitted in one of the transition is 7.5 eV. The species and the excited state ion belongs to, is

(A) H and 2nd (B) He^+ and 2nd

(C) Li^{2+} and 2nd (D) Can't be predicted

Space for rough work

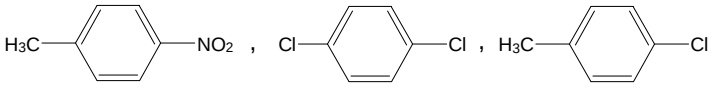
12. Which of the following is the correct order of basic nature of the compounds?



(I) (II) (III) (IV)

(A) I > II > III > IV
(B) III > I > IV > II
(C) IV > III > I > II
(D) III > IV > I > II

13. Arrange the following in order of decreasing dipole moment:



(I) (II) (III)

(A) I > II > III
(B) III > I > II
(C) I > III > II
(D) III > II > I

14. In the case of a first order reaction, the time required for 93.75% of reaction to take place is x times that required for half of the reaction. The value of 'x' is
(A) 1
(B) 2
(C) 3
(D) 4

15. In the following groups

–OAc	–OMe	$-OSO_2Me$	$-OSO_2CF_3$
(1)	(2)	(3)	(4)

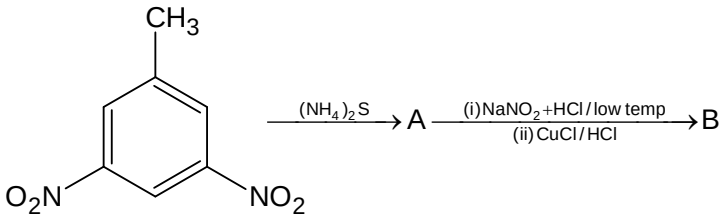
the order of leaving group ability is
(A) 1 > 2 > 3 > 4
(B) 4 > 3 > 1 > 2
(C) 3 > 2 > 1 > 4
(D) 2 > 3 > 4 > 1

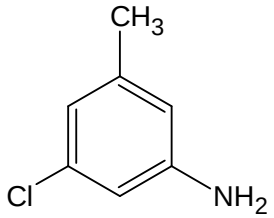
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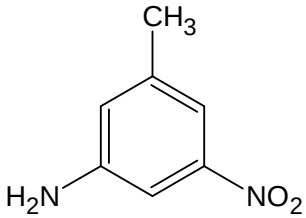
16. If 10^{-5}% reactant molecules are crossing over the barrier in transition state at 298 K, then the activation energy is

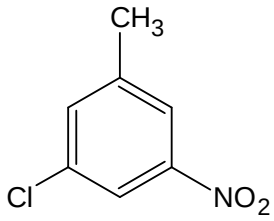
(A) 39.94 kJ (B) 49.94 kJ

(C) 79.94 kJ (D) 97.97 kJ

17. 

(A) 

(B) 

(C) 

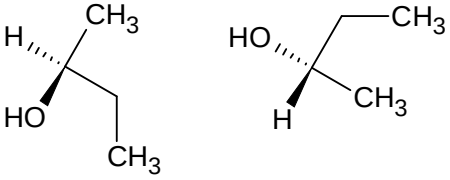
(D) None of the above

18. Which among the following compounds will give nucleophilic addition reaction?

(A) carbonyl compounds (B) nitriles

(C) isocyanides (D) all of these

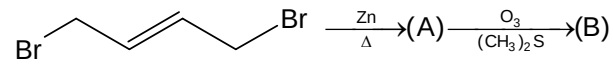
19. The pair of molecule given below represents



(A) Enentiomers (B) Diastereomers

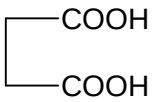
(C) Regiomers (D) None of these

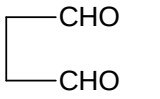
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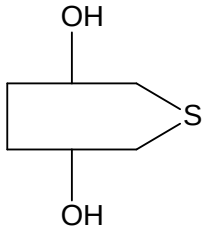
20. 

The product (A) and (B) in above sequence of reactions are

(A) (A) = $H_2C{=}C{=}C{=}CH_2$ (B) = HCHO

(B) (A) =  (B) = 

(C) (A) =  (B) = 

(D) (A) =  (B) = 

21. The hybridization of central atom in N_3^- and ICl_4^- are, respectively,

(A) sp^3d^2, sp (B) sp, sp^3d^2

(C) sp^2, sp^3d^2 (D) sp^3, sp^3d^2

22. The correct order of increasing N – O bond length in $\overset{+}{N}O_2$, NO_2^- and NO_3^- is

(A) $NO_3^- < NO_2^- < \overset{+}{N}O_2$ (B) $NO_2^- < NO_2^+ < NO_3^-$

(C) $\overset{+}{N}O_2 < NO_3^- < NO_2^-$ (D) $\overset{+}{N}O_2 < NO_2^- < NO_3^-$

23. Atoms $_7A$, $_8B$ and $_9C^{17}$ are such that $_8B$ is an isobar of $_7A$ and atom $_9C^{17}$ is isotone to B from above information. Mass number of A and B are …….. respectively.

(A) 16 and 17 (B) 16 and 16

(C) 17 and 16 (D) 15 and 16

24. Six moles of Cl_2 undergo a loss and gain of 10 mole of electrons to form two oxidation states of Cl. The difference between the maximum and minimum oxidation states of Cl in this reaction is

(A) 4 (B) 8

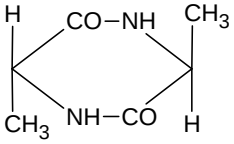
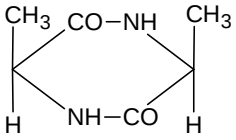
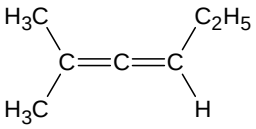
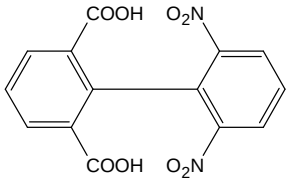
(C) 6 (D) 3

Space for rough work

25. 50 ml of 0.1 M solution of protonated form of alanine (H_2A^+, pKa_1 = 2.34, pKa_2 = 9.70) is treated with 50 ml of 0.01 NaOH. The pH of the solution will be
(A) 5.2 (B) 7.02
(C) 6.02 (D) 4.6

26. A mixture of $NH_3(g)$ and $N_2H_4(g)$ is placed in a sealed container at 27°C, 0.5 atm. After complete decomposition of NH_3 and N_2H_4 at 927°C, the total pressure reached to 4.5 atm. The percentage of NH_3 (by mole) is
(A) 75% (B) 25%
(C) 40% (D) 60%

27. When one mol of an ideal gas is compressed to half of the initial volume and simultaneously heated to twice of its initial temperature. The change in enthalpy (ΔS) is
(A) Cv ln2 (B) Cp ln2
(C) R ln2 (D) (Cv-R) ln2

28. Which of the following compounds is resolvable?
(A) 
(B) 
(C) 
(D) 

29. Borazene, $B_3N_3H_6$, is isoelectronic and isostructural with benzene. Which of the following statement(s) is/are true about borazene?
(i) Borazene is aromatic.
(ii) There are four isotopic disubstituted borazene molecules, $B_3N_3H_4X_2$.
(iii) Borazene is more reactive towards addition reactions than benzene.
(A) Only (i) (B) (i) and (ii)
(C) (i) and (iii) (D) (i), (ii) and (iii)

30. Structure of $Na_2B_4O_7.10H_2O$ contains:
(A) Two triangular and two tetrahedral units (B) Three triangular and one tetrahedral units
(C) All tetrahedral units (D) All triangular units.

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If e is the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and θ is angle between the asymptotes, then $\cos\frac{\theta}{2}$ is equal to

   (A) $\frac{1-e}{2}$ (B) $\frac{1}{e}-1$

   (C) $\frac{1}{e}$ (D) none of these

2. The sum of the abscissas of all the points on the line x + y = 4 that lie at a distance from the line 4x + 3y – 10 = 0 is

   (A) 4 (B) –4

   (C) 2 (D) –3

3. If $(x^2 + x + 2)^2 – (a – 3)(x^2 + x + 1)(x^2 + x + 2) + (a – 4)(x^2 + x + 1)^2 = 0$ has atleast one root then complete set of value of a is

   (A) $\left(1, \frac{7}{3}\right]$ (B) $\left[\frac{7}{3}, 5\right)$

   (C) $\left[\frac{3}{7}, 1\right)$ (D) $\left(5, \frac{19}{3}\right]$

4. Slope of the line $z\bar{a}+\bar{z}a+b=0$, $b \in R$ is

   (A) $-\frac{Re(a)}{Im(a)}$ (B) $-\frac{Im(a)}{Re(a)}$

   (C) $\frac{Re(a)}{Im(a)}$ (D) $\frac{Im(a)}{Re(a)}$

Space for rough work

5. If f : {1, 2, 3, 4, 5} $\rightarrow$ {x, y, z, t}, total number of onto functions 'f' is equal to
(A) 242 (B) 245
(C) 1024 (D) 240

6. Locus of mid–point of chords of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ that subtends angle $\frac{2\pi}{3}$ at the centre of the circle is
(A) $4x^2 + 4y^2 + 8gx + 8fy + 3g^2 + 3f^2 + c = 0$ (B) $4x^2 + 4y^2 – 8gx – 8fy + 3g^2 + 3f^2 + c = 0$
(C) $4x^2 + 4y^2 + 8gx – 8fy + 3g^2 + 3f^2 + c = 0$ (D) $4x^2 + 4y^2 – 8gx + 8fy + 3g^2 + 3f^2 + c = 0$

7. If the normals to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at the extremities of the chords and $px + qy = 1$ are concurrent, then
(A) $a^2 \ell p = bm^2q = –1$ (B) $a^2 \ell p = b^2mq = –1$
(C) $a \ell p = bmq = 1$ (D) $a \ell p^2 = mbq^2 = 1$

8. Let $f(x) = ax^2 + bx + c$ where $a \in R^+$ and $b^2 – 4ac < 0$. Area bounded by f(x), x–axis and the line x = 0, x = 1 is equal to
(A) $\frac{1}{6}\{3f(1)+f(-1)+2f(0)\}$ (B) $\frac{1}{12}\{5f(1)+f(-1)+8f(0)\}$
(C) $\frac{1}{6}\{3f(1)-f(-1)+2f(0)\}$ (D) $\frac{1}{12}\{5f(1)-f(-1)+8f(0)\}$

9. If $\sin^3x \sin 3x = \sum_{r=0}^{n} a_r \cos(rx)$, $\forall x \in R$ where, $a_0\ a_1\ a_2$ ….. a_n are constants and $a_n \neq 0$, then
(A) n = 5 (B) n = 6
(C) n = 7 (D) none of these

10. The numerical value of $\sum_{r=1}^{18} \sin^2(5r)^o$ is equal to
(A) 9 (B) $\frac{19}{2}$
(C) $\frac{21}{2}$ (D) $\frac{17}{2}$

Space for rough work

11. If $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then the value of $\tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left(\frac{3\sin 2x}{5+3\cos 2x}\right)$ is

(A) $\frac{x}{2}$ (B) 2x

(C) 3x (D) x

12. If a, b, c, d are positive numbers such that a + b + c + d = 2, then M = (a + b)(c + d) then M satisfies the relation

(A) $0 < M \le 1$ (B) $1 \le M \le 2$

(C) $2 \le M \le 2$ (D) $3 \le M \le 4$

13. If $1^2 + 2^2 + 3^2 + \ldots + 2003^2 = (2003)(4007)(334)$ and $(1)(2003) + (2)(2002) + (3)(2001) + \ldots + (2003)(1) = (2003)(334)(x)$, then the value of x is

(A) 2005 (B) 2004

(C) 2003 (D) 2001

14. In ΔABC, the value of the expression $\sum_{r=0}^{n} {}^nC_r a^r b^{n-r} \cos\{rB - (n-r)\}$ is equal to

(A) c^n (B) 0

(C) a^n (D) b^n

15. Total number of ways, in which 22 different books can be given to 5 students, so that two students get 5 books each and all the remaining students get 4 books each, is equal to

(A) $\frac{22!}{2!3!5!(4!)^3}$ (B) $\frac{22!}{(3!)^2\, 2!\, 5!}$

(C) $\frac{22!}{3!2!5!4!}$ (D) none of these

16. In a complex number z = x + iy such that $\left(\frac{z-1}{z+1}\right)$, $z \neq -1$ whose argument is equal to $\frac{\pi}{4}$, then relation of variables is

(A) $x^2 - y^2 = 1$ (B) $x^2 + y^2 - 2y = 1$

(C) $x^2 - y^2 + 2y + 1 = 0$ (D) none of these

Space for rough work

17. If 1, ω, ω^2 are cubes roots of unity of complex number, then $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ ..... 2n term is
(A) 0 (B) –1
(C) 1 (D) 2

18. If $i^i = e^{-(4n+1)\pi/2}$ and the value of i^i are in G.P. then the common ratio is
(A) $e^{-2\pi}$ (B) $e^{2n\pi}$
(C) e^{π} (D) e^{-1}

19. The value of $\cos\left[\tan^{-1}\left\{\sin\left(\cot^{-1}x\right)\right\}\right]$ is
(A) $\sqrt{\frac{x^2+1}{x^2+2}}$ (B) $\sqrt{\frac{x^2+2}{x^2+1}}$
(C) $\sqrt{\frac{x-2}{x^2+2}}$ (D) $\frac{x+1}{\sqrt{x^2+2}}$

20. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, and $g(x) = \left(\frac{3x+x^3}{1+3x^2}\right)$ then f[g(x)] is equal to
(A) –f(x) (B) 3f(x)
(C) $[f(x)]^3$ (D) $[f(x)]^2$

21. If $\log (1 + x + x^2 + x^3 + ...) = A \log (1 - x)$, and $A = \cos\theta$, then the value of θ is
(A) 1 (B) $\frac{\pi}{2}$
(C) π (D) 0

22. The complex number sin x + i cos 2x and cos x – i sin 2x are conjugate to each other for
(A) $x = n\pi$ (B) $x = \left(n + \frac{1}{2}\right)\pi$
(C) x = 0 (D) no value of x

23. If $z^7 + 1 = 0$ then $\cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7}$ is
(A) $\frac{1}{8}$ (B) $-\frac{1}{8}$
(C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{1}{2}$

Space for rough work

24. An ordinary cube has 4 blank faces, one face mark 2 and another marked 3, and then probability of obtaining 12 in 5 throws is

(A) $\frac{5}{2592}$ (B) $\frac{5}{1944}$

(C) $\frac{5}{1296}$ (D) $\frac{5}{648}$

25. Negation of "2 + 3 = 5" and "8 < 10" is

(A) $2 + 5 \neq$ and < 10 (B) 2 + 3 = 5 and $8 \neq <$ 10

(C) $2 + 3 \neq 5$ or $8 \neq <$ 10 (D) none of these

26. If (a, 0), (0, b) and (a, b) are the vertices of Δ circumscribed by a circle then radius of circle is

(A) $a^2 + b^2$ (B) $a^2 - b^2$

(C) $\frac{1}{2}\sqrt{a^2 + b^2}$ (D) $\sqrt{a^2 + b^2}$

27. The locus of mid–point of the chord of circle $x^2 + y^2 = y$ such that segment intersected by the chord on curve $x^2 - 2x - 2y = 0$ subtends right angles at the origin is

(A) $x^2 + y^2 - 2x - 2y = 0$ (B) $x^2 + y^2 + 2x + 2y = 0$

(C) $x^2 + y^2 = 0$ (D) none of these

28. If $C_0, C_1, C_2 \ldots C_n$ are the combinational coefficients in the expansion of $(1 + x)^n$, $n \in N$, then $C_0 + 3C_1 + 5C_2 + \ldots + (2n + 1)C_n$ is equal to

(A) $(n + 1)\, 2^n$ (B) $(n + 2)\, 2^{n-1}$

(C) $n\, 2^{n-1}$ (D) none of these

29. Let a, b, c are the sides of a triangle. No two of them are equal and $\lambda \in R$. If the roots of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real then

(A) $\lambda < \frac{4}{3}$ (B) $\lambda > \frac{5}{3}$

(C) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ (D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

30. The sum of 1st nth terms of an A.P. is given by $S_n = (1 + 2T_n)(1 - T_n)$ where T_n is nth term of series, then $T_n^2 = \frac{\sqrt{a} - \sqrt{b}}{4}$, a, b $\in$ N, then (a + b) is

(A) 4 (B) 5

(C) 6 (D) 7

Space for rough work